

# DARKSTAR

## v9 — COSMIC BOUNCE TEST

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### Executive Result

After resetting the analysis and testing the proposed cosmic rebirth directly against the Friedmann equations, the result is clear: ordinary positive-density radiation and matter do not, by themselves, create a flat-universe bounce. DarkStar therefore needs an additional dynamical ingredient if the Forked River is to reconverge into a new expanding phase.

This is not a defeat of the entire DarkStar framework. It is a sharply defined requirement. The proposed rebirth mechanism must supply either suitable spatial curvature, an effective stress-energy component with the necessary pressure behavior, modified gravity, or another mathematically explicit mechanism.

### 1. Friedmann Equation at the Turnaround

$$H^2 = (8\pi G/3)\rho - kc^2/a^2 + \Lambda c^2/3.$$

A turnaround or bounce occurs at a point where  $H = 0$ . For a spatially flat universe ( $k = 0$ ), the equation requires the total effective energy density, including any cosmological-constant contribution, to satisfy the corresponding zero- $H$  condition. Positive ordinary matter and radiation do not naturally provide that cancellation.

For positive curvature ( $k = +1$ ), the curvature term can contribute to a turnaround. A complete DarkStar model would therefore have to specify the spatial geometry rather than assuming a flat universe.

### 2. The Bounce Condition

$$\ddot{a}/a = -(4\pi G/3)(\rho + 3p/c^2) + \Lambda c^2/3.$$

At a bounce, the universe must transition from contraction to expansion, so at the minimum scale factor one requires  $H = 0$  and  $\ddot{a} > 0$ . Consequently, the total stress-energy must satisfy a sufficiently negative active-gravity combination  $\rho + 3p/c^2$ , after all terms including  $\Lambda$  are included.

Equivalently, using  $\rho_{\text{eff}} = -4\pi G(\rho + p/c^2) + kc^2/a^2$ , a flat ( $k=0$ ) bounce requires  $\rho + p/c^2 < 0$  at the bounce.

That last condition is stronger than merely having negative pressure. It points directly to a violation of the null energy condition in ordinary flat GR. Therefore a flat DarkStar bounce cannot simply be “lots of gamma radiation.” Radiation has  $p = \rho c^2/3$  and gives  $\rho + p/c^2 = 4\rho/3 > 0$ .

### 3. Radiation Fails the Bounce Test by Itself

For radiation,  $w = p/(\rho c^2) = 1/3$ . Its active-gravity combination is  $\rho(1+3w) = 2\rho$ , which is positive. Radiation therefore contributes attractive gravitational deceleration in the acceleration equation rather than the negative-pressure behavior needed for a flat bounce.

This is an important distinction for DarkStar: radiation can gravitate and can be enormously energetic without being a mechanism that reverses expansion.

### 4. What DarkStar Would Need

$$\rho_{\text{total}} = \rho_r + \rho_m + \rho_{\text{DS}} + \rho_{\Lambda}.$$

$$p_{\text{total}} = p_r c^2/3 + p_m + p_{\text{DS}} - p_{\Lambda} c^2.$$

The DarkStar component cannot simply be assigned an arbitrary value. Its density and pressure must arise from a physical model. If it behaves as  $p_{\text{DS}} = w_{\text{DS}} \rho_{\text{DS}} c^2$ , its active-gravity contribution changes sign at  $w_{\text{DS}} = -1/3$ ; however, satisfying the acceleration condition alone does not guarantee  $H = 0$ . Both equations must be solved simultaneously.

## 5. The Forked River Interpretation

The Forked River can now be translated into a mathematically meaningful sequence: dispersed state → gravitational concentration → compact-object processing → redistribution → possible global reconvergence → minimum-scale-factor state → new expansion.

The crucial scientific question is whether the reconvergence arrow can be generated by the same physics that describes the earlier branches. If not, DarkStar must identify the transition physics explicitly.

## 6. Observational Constraints

Current Planck cosmological parameter products provide constraints on matter density, curvature, the expansion rate, and other cosmological parameters. Any DarkStar bounce must reproduce these observations rather than simply fit one equation. [Planck 2018]

NASA describes the present universe as undergoing accelerated expansion attributed to an unknown dark-energy component, while noting that its physical nature remains unresolved. This leaves an open area for alternative models, but any DarkStar term must fit the observed expansion and structure data rather than merely replace the name “dark energy.” [NASA]

Recent 2026 NASA discussion also notes ongoing tests of whether dark energy may evolve with time, illustrating that the late-time expansion sector remains observationally active. [NASA 2026]

## 7. What the Reset Changed

The reset removes an important danger: we are no longer trying to make the equations confirm the original picture. The equations are being allowed to reject it.

The present result is therefore a genuine scientific checkpoint: the radiation-collapse intuition does not automatically produce a cosmic bounce. DarkStar needs a specified additional mechanism. That mechanism can now be tested rather than assumed.

## 8. Next Mathematical Target

The next step is to construct the smallest possible DarkStar effective component and derive its required equation of state, density evolution, stability conditions, and perturbations. We then test whether it can simultaneously: (1) produce a bounce; (2) avoid pathological instabilities; (3) respect appropriate conservation laws; (4) reproduce the observed hot early universe; and (5) remain consistent with CMB, expansion-history, and structure-growth constraints.

If no such component can be constructed without unacceptable contradictions, the cosmic-rebirth portion of DarkStar fails while the black-hole transformation-network portion may remain viable. If one can be constructed, its predictions become the next target of observation.

## Author and Concept Nomenclature

“DarkStar” denotes the scientific hypothesis. The author is Scott Hunter Hughes “DarkStar Aurora Shockwave.” The personal I Am philosophy is separate from this scientific model and is not used as scientific evidence.

## Illustrative equation-of-state sign test

